

Lederman's Fine-Grained Semantics

Cian Dorr | NYU | Mind and Language Seminar, 5th May 2026.

1 Scopelessness and opacitism, in natural language

In Week 1 we discussed a view of proper names as 'scopeless', which we characterized as the equivalence of all of the following:

- (1) Plato believes that Venus is visible.
- (2) a. Venus is such that Plato believes it is visible.
b. Venus is believed by Plato to be visible.
- (3) a. Something believed by Plato to be visible is identical to Venus.
b. Everything identical to Venus is believed by Plato to be visible.

The most familiar way of rejecting scopelessness is to distinguish (1) (at least on one reading) from (2a–b) and (3a–b), while treating all of the latter as equivalent to one another. This is the approach of *descriptivists* about proper names, represented in this course by Caie (2025).

The less familiar way is to treat (1) as equivalent to (2a–b), but distinguish all of these from (3a–b). Call this *opacitism*. Its natural formalization is characterized by a nonstandard logic for identity—one that lacks the standard law of substitution.

A complication: really because of the phenomenon of *contextual quantifier domain restriction*, really *no-one* should think that (3b) logically implies (2b), or that (2b) logically implies (3a), or that (3b) logically implies (3a). In a context with a narrow quantifier domain—e.g. one restricted to outer planets—(3a) will be clearly false, and (3b) will be clearly true (or at least not-false). The defensible scopelessness claim is one that says that (3a–b) are logically equivalent to (1) and (2a–b) *modulo* the truth of

- (4) Something is identical to Venus.

The interesting opacitist claim denies this.

A choice point for opacitists: are (3a) and (3b) equivalent, or not?

The *quantificationally orthodox* approach (Caie, Goodman, and Lederman 2020) says that (3b) and (4) together imply (2b), and the (2b) and (4) jointly imply (3a). Similarly, (3b) and

- (5) Hesperus is identical to something and identical to Venus

jointly imply

- (6) Hesperus is believed by Plato to be visible

and (6) and (5) jointly imply (3a). But even given (4), (3a) does not imply (2b)

or (3b), and (2b) does not imply (3b).¹

A cost of quantificational orthodoxy is having to accept the truth (in appropriate contexts) of sentences like

- (7) a. There is something believed by Plato to be visible that is identical to something not believed by Plato to be visible.

The *quantificationally unorthodox* view (Bacon and Russell 2019; Lederman 2021), by contrast, says that (3b) and (3a) are equivalent to one another, but not to (1) (or (2a–b)). Even though if *Venus*, or *Hesperus*, or *Phosphorus* is believed by Plato to be visible, it could still be that nothing identical to Venus is believed by Plato to be visible. They will reject (7). The cost they pay is having to accept sentence like (8a–b) as true (in appropriate contexts):

- (8) a. Although Venus is twinkling, and everything twinkling is believed by Plato to be visible, Venus itself is not.
b. Although Venus is twinkling and is believed by Plato to be visible, nothing twinkling is believed by Plato to be visible.

For me, this looks like a big cost! Given the admitted “delicacy” of HL's arguments against his Millian (scopelessness-endorsing) opponents, I think the balance of considerations comes out looking to favour them.

2 Higher-order Frege's puzzles

As discussed earlier in this course, English arguably contains analogues of identity in predicate and sentence position:

- (9) a. To be a groundhog is to be a woodchuck.

¹Quantificationally orthodox views include not just the 'Classical Opacity' theory of Caie, Goodman, and Lederman (2020), but also the 'Free Opacity' theories which they discuss, which characteristically reject the standard UI axiom but keep **Restricted UI**:

$$\text{UI} \quad \forall x(\phi) \rightarrow \phi[a/x]$$

$$\text{Restricted UI} \quad (\exists y(y = a) \wedge \forall x(\phi)) \rightarrow \phi[a/x]$$

Restricted quantifiers are standardly considered to obey **Restricted UI** but not UI. By contrast, the 'free opacity' theory of Bacon and Russell (2019) does *not* have **Restricted UI**, though it does have a different restriction of UI:

$$\text{Free UI} \quad \forall y(\forall x(\phi) \rightarrow \phi[y/x])$$

Free UI and **Restricted UI** are interderivable if we assume the standard substitution rule for =, but in opacity-land we don't get to do this.

- b. For there to be groundhogs on Hesperus is for there to be woodchucks on Phosphorus.
- c. For it to be the case that Ortcutt is a spy and Ortcutt is loyal is for it to be the case that Ortcutt is a loyal spy.

These generate analogues of Frege’s puzzle. Although HL doesn’t discuss these, it’s straightforward to extend his ‘Basic Surrogatist’ and ‘Enriched Surrogatist’ views to them.

- HL’s models come equipped with a distinguished accessibility relation R , which doesn’t do anything in Basic Surrogatism, but in Enriched Surrogatism provides a constraint on the allowable values of the function f which constrains the existential quantification over permutations. We can also use this R to give a semantics for higher-order identity: say that ‘to be F is to be G ’ is true at w iff for all $x \in D_e$ and all $w' \in R(W)$, $\llbracket F \rrbracket(x)(w') = \llbracket G \rrbracket(x)(w')$. This will allow for higher-order opacity: e.g., ‘Joe believes Phil is a groundhog, but does not believe Phil is a woodchuck, although to be a groundhog is a woodchuck’ can be true (in a model where $\llbracket \text{groundhog} \rrbracket \neq \llbracket \text{woodchuck} \rrbracket$). This seems like what one would expect.
- If he wanted to include higher-order quantification in the language (e.g. as a treatment of ‘some properties...’ and ‘every property...’), HL could impose a parallel constraint to the one he endorses in the first order case: the domain for a world must contain only one element of each equivalence class under the denotation of “identical” at that world.
- The apparatus of permutation variables characteristic of Enriched Surrogatism can be extended smoothly to other types. We can define ‘permutation function admissible at w' ’ for an arbitrary domain D_σ in the same way that HL defines it for D_e . And in a model, we let the value of f relative to a world and an agent be a *typed collection* of permutation functions admissible at that world, nonempty in each type.
- HL suggests, but does not exactly argue for, the constraint that the identity permutation always must belong to every $f(x)(w)$. If we drop this, Enriched Surrogatism extended to higher types also offers a solution to the *problem of logical omniscience*: we can say, e.g., that ‘not’ has denotes the function $\lambda p. \lambda w. 1 - p(w)$ in $D_{(s \rightarrow t) \rightarrow s \rightarrow t}$, and that in certain contexts, the value of $f(\text{Graham Priest})(@)$ contains only permutations of this domain such that when w is any of Priest’s belief-worlds, there are sets of worlds p such that $w \in p$ and $w \in n(p)$.

In formal higher-order languages, we have predicate variables which can be bound by λ , so we get analogues of (2a–b) for predicates: $(\lambda X. B(Xa))F$ versus $B(Fa)$. It’s a bit awkward, but I think possible, to get the effect of this in natural language:

(10) Loving Hesperus is such that Joe thinks Plato does it.

One might expect that the higher-order extensions of Surrogatism would make

(10) equivalent to

(11) Joe thinks Plato loves Hesperus

analogous to how (1) comes out equivalent to (2a–b). But because of the apparatus of permutation variables, this is not the case. in (11), we can have three permutation variables, one each on ‘Plato’, ‘loves’, and ‘Hesperus’, while with (10), we’ll presumably just get two: one on ‘Plato’ and one on ‘does it’ (which we are treating as functioning like a preedicate variable). This can make for differences.

This, arguably, somewhat diminishes the dialectical advantages of being able to treat (1) and (2a–b) as equivalent.

3 Metaphysical reductions

HL suggests that those who want ‘a semantics in the philosophers’s heavy-weight sense’ should say that what proper names *denote* [*simpliciter*] are not individuals, but something else, which might be called a ‘way of thinking of an individual’.

I think what HL is envisaging here is a theory semantics (not just model-theoretic ‘semantics’) for English stated in a formal language with a fully standard (classical) logic. Such a semantics will also induce a non-obvious *translation* from English into such a language—one on which English sentences which are counterexamples to Substitution will be translated into sentences of the formal languages that are consistent with its version of Substitution.

Here is a ‘two-dimensionalist’ picture of how this might go, using an extensionalist HO logic with three basic types s (‘ur-worlds’); e (‘ur-individuals’), and b (‘Boolean truth values’).

- English proper names have denotations in type $s \rightarrow e$. (They stand for ‘individual concepts’, generally non-constant ones).
- English sentences have denotations in type $s \rightarrow s \rightarrow b$. (They stand for ‘individual concepts of ur-propositions’, or ‘two-dimensional propositions’.) The role of “worlds” in HL’s semantics is thus played by ordered pairs of ur-worlds; the R relation is ‘having the same first co-ordinate’.
- General monadic predicate denotations are of type $(s \rightarrow e) \rightarrow s \rightarrow s \rightarrow b$. But a *transparent* predicate is associated with some $F : s \rightarrow e \rightarrow s \rightarrow b$ (an ‘individual concept of ur-properties’): its denotation is $\lambda x^{s \rightarrow e} w. Fw(xw)$.
- ‘Identical’ denotes $\lambda x^{s \rightarrow e} y^{s \rightarrow e} ww'. (xw) = y(w)$.
- ‘Believes’ is of type $(s \rightarrow e) \rightarrow (s \rightarrow s \rightarrow b) \rightarrow s \rightarrow b$. But we can (if we wish) analyse it in terms of an underlying DOX of type $e \rightarrow (s \rightarrow b) \rightarrow s \rightarrow b$: it denotes $\lambda x^{s \rightarrow e} p^{s \rightarrow s \rightarrow b}. \text{DOX}(xw)(\lambda v. pvv)$.

I don’t want to get too bogged down in the details of this. But I find it metaphysically confusing, because I feel like if I can understand this formal lan-

guage, I should be able to introduce into English a predicate ‘basic’, such that $\llbracket \text{is basic} \rrbracket(x)(w)(w') = 1$ iff for some y^e , $x = \lambda v.y$. I can then ask questions like ‘Is Hesperus basic?’ ‘Is Vebus basic?’ ‘Is there anything basic identical to Venus?’

And maybe I can get myself into a context where ‘Everything is basic’ is true (using ‘basic’ as the manadory quantifier-restrictor for ‘everything’), and in this context ask questions like ‘Is anything identical to Venus believed by Plato to be bright?’

The questions seem hard! I guess the least arbitrary thing to say is that typical English names don’t stand for basic things, and people don’t directly (i.e. on a permutation-variable-free raeding) believe very much about basic things.

But I find something fundamentally mysterious in the picture. Does anything stop me from introducing a new name by saying ‘Let Schmesperus be the basic thing identical to Hesperus?’ Having done this, doesn’t it seem like ‘I know that Schmesperus is identical to Hesperus’ should be true? But what if I also introduce ‘Schmosphorus’ in the same way...?

4 Shared problems for Harvey and CG theory

Lots of structural ambiguity: we can introduce the variables or leave them out, and we can use the same variable on different occurrences of names, or different ones. Can this all be motivated?

The problem of constraint: do we have the option of stacking permutation variables on top of each other? In a nested attitude report, there seem to be constraints on the order of binding and world-variable-saturation; how can they be explained? What about in sentences with regular (circumstantial) modal operators under or over attitude reports?

The problem with polyadic attitude verbs: how do we stop ‘Ralph is more confident that Ortcutt is a spy than that Ortcutt is a spy’ from having true readings?²

The problem of quantification over objects-of-attitudes: Consdier arguments like the following

- (12) a. She believes everything the Pope believes.
- b. He believes everything she wanted him to believe.

If these are just taken as quantification over sets of worlds, we don’t get inferences like

- (13) Ann believes that Venus is visible.
- Bea believes that Venus is visible.

²One way around this is to go for the ‘disjunctive’ approach briefly discussed in my session. Instead of using existential quantification over permutations, take the disjunction of all the propositions generated by values of the permutation variables that satisfy the f constraint, and see if that disjunction holds in all the person’s belief worlds.

Cat does not believe that Venus is visible.

Therefore, there is something that Ann and Bea believe that Cat does not believe.

Even if we stick with what HL calls an ‘insensitive’ theory (where the domain-giving function f does not vary across individuals), the truth of the premises is still consistent with Cat’s belief-worlds being the union of Ann’s and Bea’s.

To avoid this, we might think that the quantification is over the type of the argument of ‘believe’ (functions from tuples of permutation functions to worlds to truth values). The problem is that there is no single *type* that contains all the different arguments that ‘believe’ can take. And in the higher-order generalization, there’s no way to represent all the possible kinds of arguments within a single type.

5 Harvey’s argument against the the Indexed-Domain CG theory

On this theory (as on mine), switching between different names for the same object merely serves to suggest different resolutions for a context-sensitive element (in the CG theory: a set of concept-generators); nothing in principle stops this from happening even when it’s the same name.

HL notes that it’s extremely hard to get the relevant readings for sentences with repeated occurrences of the same name:

- (14) Amalia didn’t know Issa had any relevant genes in common with Issa.

I agree that this is hard in the context of HL’s example, and I don’t know how best to explain this. But I think there are other contexts where (14) can be good (even though the two occurrences of ‘Issa’ would still be regarded by Harvey as co-denoting). Suppose Amalia is a researcher who never met Issa but has been sequencing Issa’s genome, and knows her name. She’s at party where she can see some people obviously do share genes from the genome, and others don’t. She is introduced to Issa, but not by name, and because the lighting situation is bad she can’t see any of the halllmarks of the sequenced genome in Issa’s appearance.

References

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